Thermo Review

First Law - Conservation of Energy

$KE = \frac{1}{2}mV^2 \rightarrow \frac{KE}{m} = \frac{1}{2}V^2$

- extensive property
- specific property

$Q = C\Delta T$ ← temperature

- material coefficient

C_v - constant volume - use this

C_p - constant pressure

- potential energy; pressure - $\frac{PE}{m} = PV = \frac{P}{\rho}$

$\rightarrow \frac{E}{m} = \frac{1}{2}V^2 + C_v T + \frac{P}{\rho}$

- all dimensions $\left[\ell^2/t^2\right]$

· Adiabatic - no heat transfer through system
- system energy conserved

Enthalpy - thermal and potential combined
- thermal + pressure potential $h = E_{int} + PV$

Ideal Gas Law - $pV = nRT \rightarrow p = \frac{n}{V}RT \rightarrow p = \rho RT$
- $R = 8.314/5 \frac{J}{K \cdot mol}$ $R_{air} = 287.05 \frac{J}{kg \cdot K}$

$\frac{1}{2}U^2 + \frac{P}{\rho} + C_v T = \frac{1}{2}U^2 + RT + C_v T = \frac{1}{2}U^2 + C_p T$, $C_p = C_v + R$

$C_{v,air} = \frac{5}{2}R$ $C_{p,air} = \frac{7}{2}R$ $\gamma = \frac{C_p}{C_v} = \frac{7}{5}$

· If hypersonic, $C_v \uparrow$ but ignore for this course

$e_{int} = C_v T$ $h = C_p T$

First Law + adiabatic → stagnation (total) temp

$\frac{1}{2}U^2 + \frac{P}{\rho} + C_v T = \text{const} \rightarrow \frac{1}{2}U^2 + C_p T = C_p T_0$

$\rightarrow U^2 = 2C_p(T - T_0) \rightarrow \frac{T_0}{T} = 1 + \frac{U^2}{2C_p T}$

Ex. Air accel from rest, temp ↓ by 10K

$U = \sqrt{2C_p(T_0 - T)} = \sqrt{2 \cdot 1004 \cdot 10} =$

2nd Law - Entropy - Heat flows from hot to cold

- entropy ↑ during heat transfer / friction
- isentropic - not heat loss / entropy, $\Delta s = 0$

$C_v dT = -p\frac{dV}{dM} \rightarrow -p d\frac{1}{\rho} = C_v dT \rightarrow \frac{d\rho}{\rho} = \frac{C_v dT}{RT}$

$\rightarrow \frac{\rho_0}{\rho} = \left(\frac{T_0}{T}\right)^{C_v/R} \qquad \frac{P_0}{P} = \left(\frac{T_0}{T}\right)^{C_p/R}$

stagnation $P_0 = \rho_0 R T_0$ static $P = \rho R T$

- P_0 does not change in isentropic flow

$\frac{P_0}{P} = \frac{\rho_0}{\rho}\frac{T_0}{T} = \left(\frac{T_0}{T}\right)^{C_v/R+1} \qquad \frac{C_v}{R+1} = \frac{C_v + R}{R}$

Ex. tank at $P_0 = 10\text{atm}$, $T_0 = 500K$ exhausts to $P = 1\text{atm}$, find T, isentropic

$\rightarrow T = T_0\left(\frac{P}{P_0}\right)^{2/7} = 500(0.1)^{2/7} = 259K$

Isentropic Flow

· First Law, adiabatic $\frac{T_0}{T} = 1 + \frac{U^2}{2C_pT}$

· isentropic (inviscid)

$$\frac{P_0}{P} = \left(\frac{T_0}{T}\right)^{C_p/R} = \left(1 + \frac{U^2}{2C_pT}\right)^{C_p/R}$$

$$\frac{\rho_0}{\rho} = \left(\frac{T_0}{T}\right)^{C_v/R} = \left(1 + \frac{U^2}{2C_pT}\right)^{C_v/R}$$

Ex. Airfoil in isentropic flow: $P_\infty = 1\,atm$, $T_\infty = 267K$, $U_\infty = 100\,m/s$

$\rightarrow P_0 = 1.067\,atm$, $T_0 = 271.97K$

on airfoil, $p = 0.7 \rightarrow T = T_0\left(\frac{p}{P_0}\right)^{2/7} = 271.97(0.7/1.067)^{2/7} = 241.1K$

$\rightarrow U^2 = 2C_p(T_0 - T) \rightarrow U = 249\,m/s$

Entropy change $s_2 - s_1 = c_p \ln\frac{T_2}{T_1} - R\ln\frac{P_2}{P_1} = c_v \ln\frac{T_2}{T_1} + R\ln\frac{V_2}{V_1}$

- if adiabatic, $T_0 = const \rightarrow \Delta s = -R\ln\left(\frac{P_{02}}{P_{01}}\right)$ isentropic: $P_0 = const$

where T_0, P_0 = total (stagnation)

upstream		downstream	
$P_\infty, T_\infty, V_\infty$		$P_2 = 0.7$	
P_{01}	=	P_{02}	isentropic
T_{01}	=	T_{02}	- adiabatic

$T_0 = T + \frac{V^2}{2c_p}$

$\frac{P_0}{P} = \left(\frac{T_0}{T}\right)^{C_p/R} = \left(\frac{T_0}{T}\right)^{7/2}$

· Shock - not isentropic! $P_{02} \leq P_{01}$

Compressible Flow

- can the gas be compressed by the flow alone
  - ρ decrease due to flow

$$\frac{\rho_0}{\rho} = \left(1 + \frac{V^2}{2c_pT}\right)^{5/2}$$

Speed of Sound

- $V \ll 1$ - inviscid
- displacement + restoring force
  - more compressible, less restoring force → slower speed

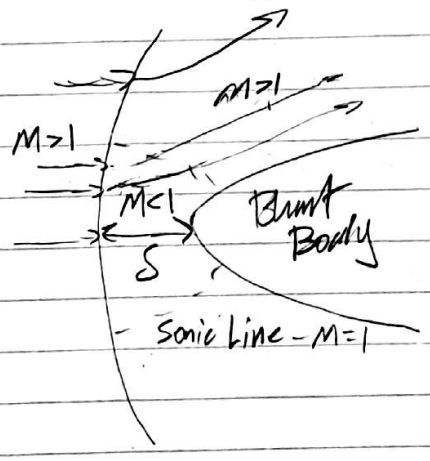
$$a^2 = \gamma RT \qquad M = \frac{V}{a} = \frac{V}{\sqrt{\gamma RT}} \qquad a = \sqrt{\frac{dP}{d\rho}}$$

$$\frac{T_0}{T} = 1 + \frac{U^2}{2c_pT} = 1 + \frac{U^2}{a^2}\frac{\gamma R}{2c_p} = 1 + \frac{\gamma R}{2c_p}M^2 \qquad \gamma = \frac{c_p}{c_v},\ c_v = \frac{5}{2}R$$

$$= 1 + \frac{R}{2c_v}M^2 \rightarrow \frac{T_0}{T} = 1 + \frac{1}{5}M^2 \rightarrow M = \sqrt{5\left(\frac{P_0}{P}\right)^{2/7} + 1}\ ?$$

$M > 1$, Shock

- flow straight until it reaches shock



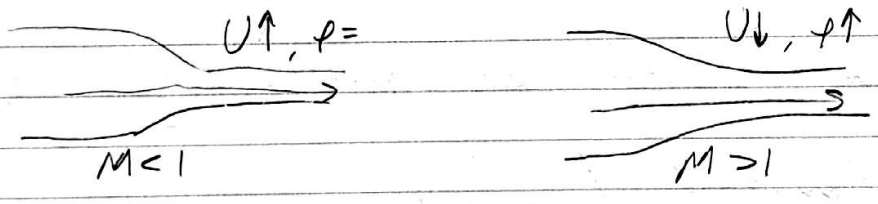
Isentropic Flow Formulas

· In terms of Mach number

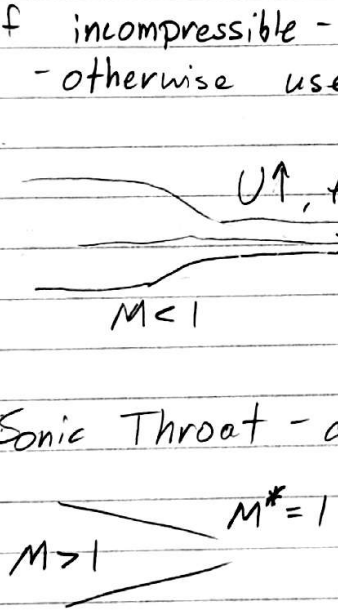
$$\frac{U^2}{2C_pT} = \frac{(\gamma-1)}{2}M^2 \qquad \frac{T_0}{T} = \left(1+\frac{\gamma-1}{2}M^2\right)$$

$$\frac{P_0}{P} = \left(1+\frac{1}{5}M^2\right)^{7/2} \qquad \frac{\rho_0}{\rho} = \left(1+\frac{1}{5}M^2\right)^{5/2}$$

· if incompressible - Bernoulli's eqn works fine
- otherwise use Mach eqns



· Sonic Throat - adiabatic - not isentropic



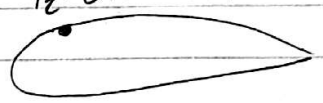
this causes problems

· for T^*, P^*, ρ^*, plug in $M=1$

Ex.

$M_1 = 0.6$
$P_1 = 1.1\text{atm}$

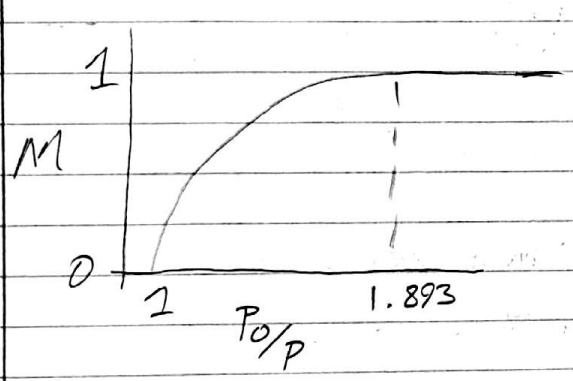
$P_2 = 0.85\text{atm} \rightarrow M_2?$



- assume isentropic
$\rightarrow P_{0_1} = P_{0_2}$

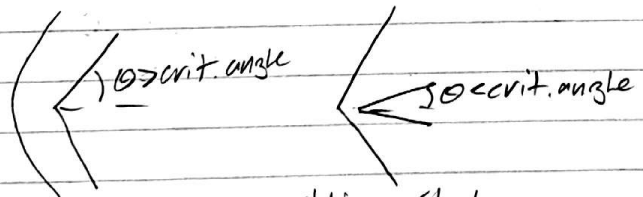
$$\frac{P_2}{P_1} = \frac{P_2}{P_0}\frac{P_0}{P_1} = \left(\frac{1+\frac{1}{5}M_1^2}{1+\frac{1}{5}M_2^2}\right)^{7/2}$$

Choking / Regulating



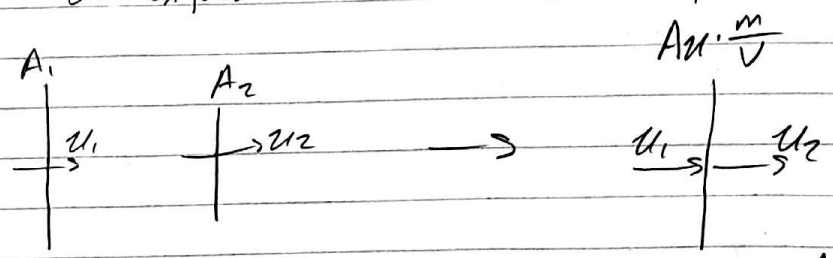
Shocks

- Inviscid flow - no viscosity - no shock thickness - otherwise $t = \frac{\nu}{a}$



Normal Shock (Detached) — Oblique Shock (Attached)

- much behind normal shock < 1, much behind oblique > 1
- form inside nozzles - needs large / quick pressure increase to adapt to ambient pressure - gas before shock overexpands and loses too much pressure



$A_1 u_1 \Delta t = A_2 u_2 \Delta t$

volumetric flow rate

$\rho_1 u_1 A_1 = \rho_2 u_2 A_2$

$\rightarrow A_1 = A_2 \rightarrow \rho_1 u_1 = \rho_2 u_2$ Mass consv.

'consv. of Energy':

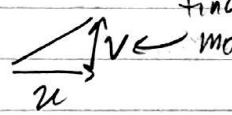
$$T_1 + \frac{1}{2}\frac{u_1^2}{c_p} = T_{o_1} = T_{o_2} = T_2 + \frac{1}{2}\frac{u_2^2}{c_p}$$

- ideal gas on either side of shock: $p_{1,2} = \rho_{1,2} R T_{1,2}$

'consv. of momentum'

$\rho_1 u_1^2 \neq \rho_2 u_2^2$ - pressure difference across shock causes additional force backwards

$mom_2 - mom_1 = F = p_1 - p_2 \rightarrow p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2 \leftarrow$ normal



find tangential momentum $\rightarrow$ P const. $\rightarrow \rho_1 v_1 u_1 = \rho_2 v_2 u_2$ tangential

$\rightarrow v_1 = v_2 \leftarrow$

$\rightarrow$ always: $\frac{m}{\Delta t}$ in $= \frac{m}{\Delta t}$ out

Normal Shocks

$M_1 > 1$, $M_2 < 1$

$$M_2^2 = \frac{5 + M_1^2}{7M_1^2 - 1} \geq \frac{1}{7} \rightarrow M_1^2 = \frac{5 + M_2^2}{7M_2^2 - 1}$$

$$\frac{u_1}{u_2} = \frac{\rho_2}{\rho_1} = \frac{6M_1^2}{5 + M_1^2} \qquad \frac{P_2}{P_1} = \frac{7M_1^2 - 1}{6}$$

$$\frac{T_2}{T_1} = \frac{(7M_1^2 - 1)(5 + M_1^2)}{36M_1^2} \rightarrow \text{loss: } \frac{P_{o_2}}{P_{o_1}} = \frac{P_2}{P_1}\left(\frac{T_1}{T_2}\right)^{7/2}$$

$P_{o_1} > P_{o_2}$ - 2nd Law

- between shock and stagnation point is isentropic

$$S_2 - S_1 = -R \ln \frac{P_{o_2}}{P_{o_1}}$$

- across shocks are adiabatic $\rightarrow T_{o_1} = T_{o_2}$

$$\frac{T_2}{T_1} = \frac{\frac{P_2}{P_1}}{\frac{\rho_2}{\rho_1}}$$

Pitot Tube Formulas

Subsonic: $M_1^2 = \frac{2}{\gamma-1}\left[\left(\frac{P_{01}}{P_1}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]$

Supersonic: $\frac{P_{02}}{P_1} = \left[\frac{(\gamma+1)^2 M_1^2}{4\gamma M_1^2 - 2(\gamma-1)}\right]^{\frac{\gamma}{\gamma-1}} \left[\frac{1-\gamma+2\gamma M_1^2}{\gamma+1}\right]$

$$= \frac{6^6 M_1^7}{5^{7/2}(7M_1^2-1)^{5/2}} \rightarrow \frac{6^6 M_1^7}{5^{7/2}(7M_1^2-1)^{5/2}} - \frac{P_{02}}{P_1} = 0$$

- scipy.optimize -> fsolve -> find roots of M_1
(Newton's Method?)

$\frac{P_{02}}{P_1}$ must be > 1.8929 -> value at Mach = 1